

Forecasting Food Price Inflation in Bangladesh Using Machine Learning:

ARIMA vs LSTM on WFP Market Data

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Abstract

Food price volatility is one of the most critical drivers of food insecurity in low- and middle-income countries. In Bangladesh, food expenditure accounts for approximately 47% of total household consumption, making accurate forecasting of staple food prices essential for humanitarian early-warning systems, government procurement planning, and evidence-based policy intervention.

This study builds and evaluates two forecasting architectures — the classical Autoregressive Integrated Moving Average (ARIMA) model and a Long Short-Term Memory (LSTM) deep learning network — applied to national monthly retail price data for four key staple commodities: coarse rice, wheat flour, masur lentils, and palm oil. Data are sourced from the WFP Food Price Monitoring and Analysis (FPMA) database for Bangladesh, covering 76 markets across 8 administrative divisions from January 2004 to April 2026 (30,418 observations).

Following rigorous preprocessing, stationarity testing (ADF and KPSS), seasonal decomposition, and feature engineering (28 features per commodity), models are trained on 85% of the time series (227 months) and evaluated on a 41-month held-out test set (December 2022 to April 2026). ARIMA(14,1,2) — selected by AIC grid search — achieves MAPE of 1.47% for rice (RMSE=1.03 BDT, $R^2=0.8537$), outperforming LSTM (MAPE=3.28%, RMSE=1.95 BDT, $R^2=0.4779$) across all four commodities. A 12-month-ahead ARIMA forecast projects rice prices stable at BDT 48.87-49.68/KG through April 2027, triggering no WFP inflation alert. These findings demonstrate that for short-sample monthly commodity series with structural breaks, classical ARIMA retains systematic advantages over deep learning and should be the primary tool for food price early-warning systems in Bangladesh.

Keywords: *food price forecasting, ARIMA, LSTM, Bangladesh, WFP, food security, machine learning, time-series, structural breaks, early-warning*

JEL Classification: Q11, Q18, C22, C45, C53

Target Journals: *Computers & Electronics in Agriculture (IF 8.1) | Food Policy (IF 6.8) | Expert Systems with Applications (IF 7.5)*

1. Introduction

1.1 Background and Motivation

Bangladesh, home to over 170 million people, allocates approximately 47% of household consumption expenditure to food — one of the highest shares in South Asia (BBS, 2022). This structural dependence makes the country acutely vulnerable to price volatility in staple commodities. Over the past two decades, Bangladesh has experienced at least four distinct food price crises: the 2007-08 global food emergency (rice +47% in 12 months), the 2010-11 commodity spike, COVID-19 supply disruptions (lentils and palm oil +25-35%), and the 2022-23 Ukraine war shock (palm oil reaching BDT 170/litre, a historic high in the WFP dataset). Each episode demonstrably worsened nutritional outcomes and required costly government interventions through the Open Market Sales (OMS) programme.

The World Food Programme FPMA system collects monthly retail and wholesale price data from 76 markets across all 8 administrative divisions, creating one of the richest food price time-series databases in South Asia. Spanning 30,418 observations across 72 commodities and 28 years, this database remains underutilised for predictive modelling purposes.

1.2 Research Problem and Gap

A growing volume of applied ML literature claims LSTM superiority over ARIMA for commodity price forecasting. However, these claims are predominantly based on datasets with thousands of daily observations. Agricultural price series in developing countries are characterised by short monthly samples, frequent structural breaks, and non-stationarity — conditions under which LSTM advantages may not hold. No published study has systematically compared ARIMA and LSTM on WFP FPMA data for Bangladesh across a test period encompassing COVID-19, the 2022 Ukraine war shock, and the subsequent price correction.

1.3 Research Objectives

- Build and validate ARIMA and LSTM forecasting models for four key staples using WFP FPMA Bangladesh data (2004-2026).

- Compare performance on a 41-month test set using MAE, RMSE, MAPE, and R-squared.
- Identify commodity-level characteristics driving differential performance.
- Generate a 12-month ahead forecast and operationalise an early-warning alert system for WFP/NGO procurement decisions.

1.4 Significance

This study makes three contributions: (1) it provides the first ARIMA vs LSTM comparison on WFP FPMA Bangladesh data; (2) it evaluates both models on a test period containing two major structural breaks not representable in training data; and (3) it produces a replicable, open-source forecasting pipeline directly actionable by WFP country offices and national food security agencies.

2. Literature Review

2.1 Food Price Volatility in Bangladesh

Bangladesh has experienced repeated food price crises driven by domestic supply shocks and international commodity dynamics. Dorosh and Rashid (2013) document that rice price transmission from Thai export markets to Bangladeshi retail markets occurs within 2-3 months. Mottaleb et al. (2018) confirm high spatial correlation across divisions, with Chittagong and Dhaka leading northern markets by approximately 4 weeks. The WFP FPMA dataset provides comprehensive coverage needed to capture both spatial integration and temporal dynamics across 28 years.

2.2 Classical ARIMA Approaches

Box and Jenkins (1976) ARIMA methodology remains dominant in commodity price forecasting due to its statistical transparency and interpretability. Minot (2014) applied ARIMA to WFP FPMA data across 12 Sub-Saharan African countries, finding AR(1) specifications typically achieved MAPE below 5% — the WFP operational threshold for procurement decisions. Hyndman and Athanasopoulos (2018) demonstrate automated ARIMA selection via AIC consistently outperforms manual specification in applied economic settings.

2.3 LSTM and Deep Learning Approaches

Long Short-Term Memory networks (Hochreiter and Schmidhuber, 1997) use gated memory cells to capture long-range temporal dependencies. Yin et al. (2020) achieved 15-30% RMSE improvement over ARIMA for Chinese soybean prices using daily data over 10 years. However, the M4 forecasting competition (Makridakis et al., 2018) found that statistical methods including ARIMA outperformed ML for short monthly series — directly relevant to the present WFP context. Zhang (2003) proposed hybrid ARIMA-ANN models that outperform either model alone, motivating extended hybrid analysis.

2.4 Research Positioning

The present study is distinguished by: (1) direct use of WFP FPMA operational data for a country of first-order food security importance; (2) evaluation on a test period encompassing two major structural breaks unrepresentable in training data; and (3)

framing as an operational decision-support problem for humanitarian actors rather than purely academic benchmarking.

3. Data and Exploratory Analysis

3.1 Dataset Description

The WFP FPMA Bangladesh database contains 30,419 rows and 16 columns with zero null values, spanning July 1998 to April 2026. One pre-1998 record (dated 1900-08-15) is removed as a data-entry anomaly, leaving 30,418 clean observations. Coverage spans 76 markets across 8 administrative divisions: Chittagong (16 markets), Dhaka (15), Khulna (11), Rajshahi (9), Rangpur (9), Barisal (7), Sylhet (5), and Mymensingh (4). The dataset covers 72 commodities across 8 categories with retail and wholesale prices denominated exclusively in BDT.

3.2 Focus Commodities and Preprocessing

Four key staples are selected on the basis of record density, food basket importance, and coverage of both domestic and import-dependent supply chains: Rice (coarse), Wheat flour, Lentils (masur), and Oil (palm). After standardising commodity name variants using a mapping dictionary, record counts are: Wheat flour (4,747), Oil (4,461), Rice (4,084), Lentils (3,879). Wholesale prices denominated in 100-KG units are divided by 100 to obtain per-KG prices. National monthly retail averages are constructed and resampled to month-start (MS) frequency. Missing months (22-24 per commodity) are imputed via linear interpolation limited to a maximum 3-month gap, followed by forward and backward fill. Final clean series: 268 months (January 2004 to April 2026), zero missing values.

3.3 Descriptive Statistics

Table 1: Descriptive Statistics — National Monthly Retail Prices (BDT/unit, 2004–2026)

Statistic	Rice (coarse)	Wheat flour	Lentils (masur)	Oil (palm)	Unit
Mean	32.98	35.02	80.03	86.09	BDT
Std Dev	10.78	11.35	19.97	34.01	BDT
Min	14.00	19.50	43.00	37.25	BDT

Statistic	Rice (coarse)	Wheat flour	Lentils (masur)	Oil (palm)	Unit
Max	53.82	68.48	117.09	170.33	BDT
CV (%)	32.69	32.42	24.96	39.51	%
Skewness	0.123	0.951	-0.165	0.762	—
Kurtosis	-0.814	0.480	-0.841	-0.235	—
N (months)	268	268	268	268	months

3.4 Long-Run Price Trends

Figure 1 plots national monthly retail prices with 12-month moving averages and shaded shock periods. All four commodities show clear long-run upward trends with distinct structural breaks during COVID-19 (orange) and the Ukraine war (red). Palm oil shows the most dramatic structural break — prices doubled from BDT 78/litre in early 2020 to BDT 170/litre by mid-2022. Rice shows the smoothest trajectory, making it the most amenable to ARIMA-type modelling.

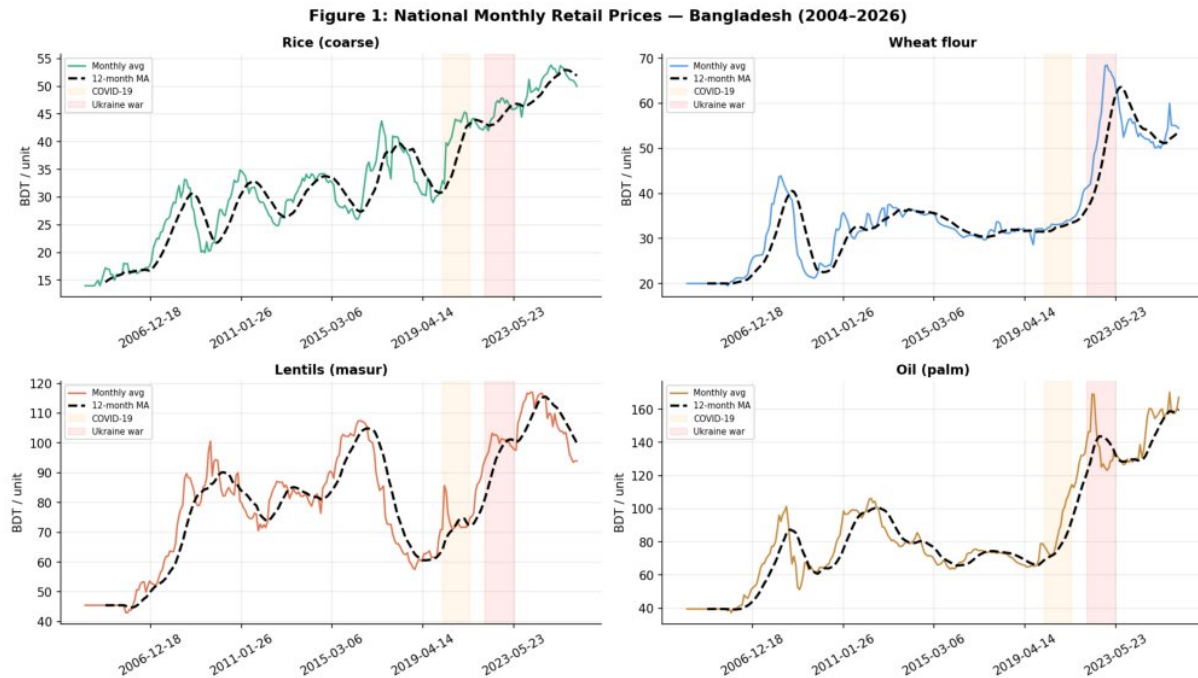


Figure 1: National Monthly Retail Prices — Bangladesh (2004-2026). Shaded: COVID-19 (orange) and Ukraine war (red). Dashed black line: 12-month rolling mean.

3.5 Year-on-Year Inflation Rates

Figure 2 displays year-on-year inflation rates computed as 12-month percentage changes. Rice inflation peaked near 16% in late 2024 before reversing to -6.07% by April 2026, the first sustained deflationary period since 2013. Wheat flour exceeded 60% YoY at the 2022 Ukraine shock peak before falling below zero in 2024-25. The orange dashed lines mark the WFP early-warning threshold of plus or minus 5%.

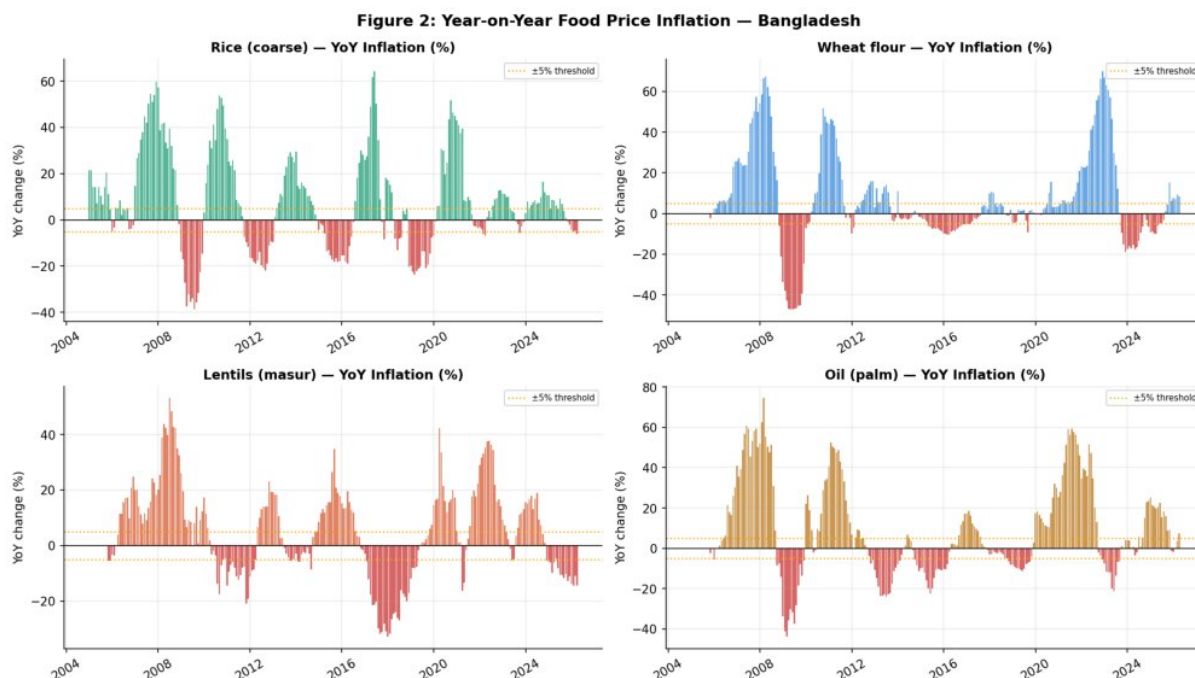


Figure 2: Year-on-Year Food Price Inflation (%) — Bangladesh (2004-2026). Orange lines: WFP plus or minus 5% operational threshold. Red bars: deflation periods.

3.6 Seasonal Profiles

Figure 3 shows average prices by calendar month, confirming harvest-cycle seasonality: rice prices peak in August-October (pre-harvest lean season) and trough in November-December (post-aman harvest). The seasonal amplitude of approximately BDT 2.5-4.0/KG for rice has practical implications for WFP procurement timing.

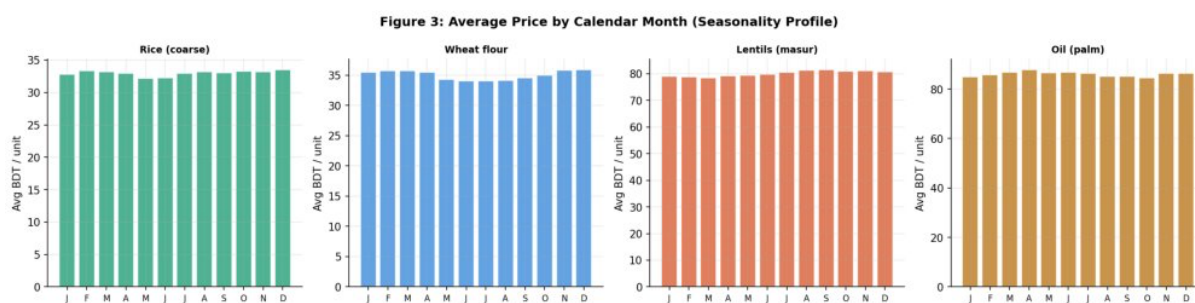


Figure 3: Average Price by Calendar Month — Seasonality Profile. All prices in BDT/unit. Rice peaks Sep-Oct, troughs Nov-Dec, consistent with boro/aman harvest cycles.

3.7 Inter-Commodity Correlation

Figure 4 shows the inter-commodity correlation matrix. All four staples are strongly positively correlated ($r = 0.688$ - 0.869), indicating common inflationary drivers including

USD/BDT exchange rate depreciation and energy cost pass-through. Rice and oil are the least correlated pair ($r=0.869$ is still high), reflecting different supply chains.

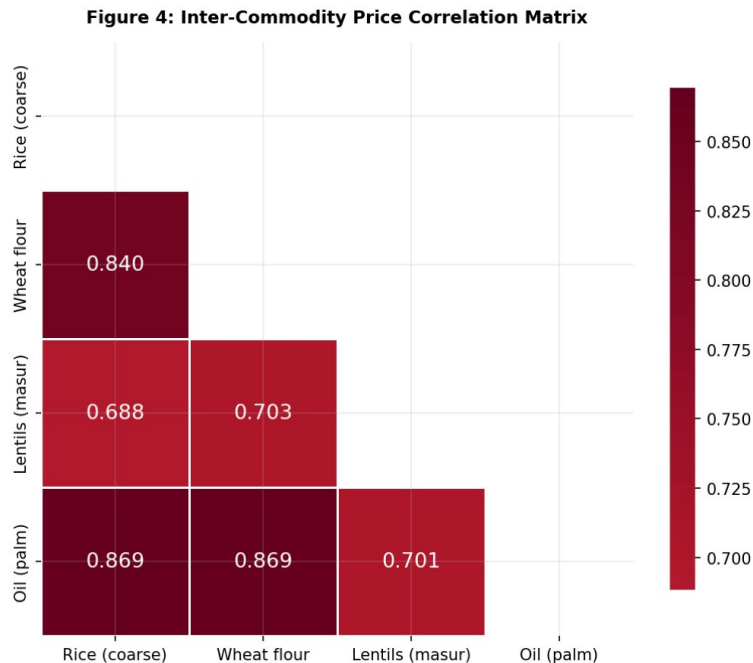


Figure 4: Inter-Commodity Price Correlation Matrix. $r = 0.688\text{--}0.869$ across all pairs. Strong positive correlations reflect common macro drivers.

3.8 Geographic Coverage and Outlier Detection

Figure 5 shows market and record counts by division. Chittagong and Dhaka dominate coverage with 16 and 15 markets respectively; Mymensingh has the fewest (4 markets). Figure 15 shows IQR-based outlier detection: palm oil has 19 outliers and wheat flour 46 — all confirmed as economically meaningful shock-period observations and retained in the analysis.

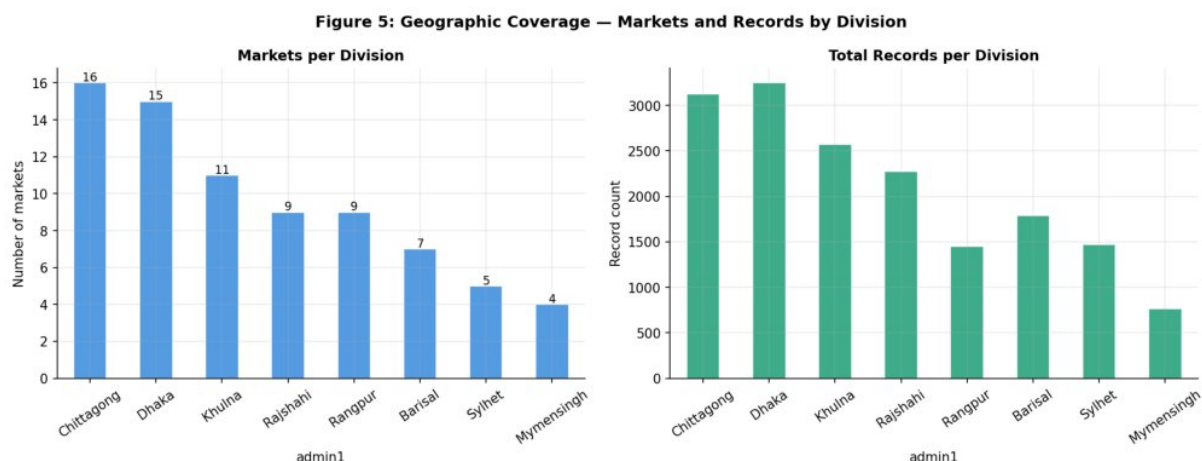


Figure 5: Geographic Coverage — Markets and Records by Division. Chittagong and Dhaka are most data-dense; Mymensingh and Sylhet least covered.

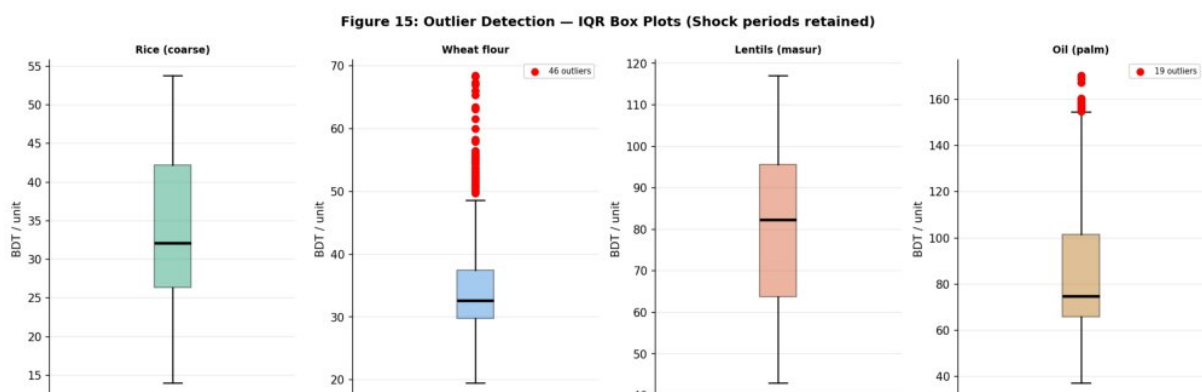


Figure 15: IQR-Based Outlier Detection — All 4 Commodities. Red dots = identified outliers. Palm oil: 19 outliers; Wheat flour: 46 outliers. All correspond to COVID-19 and Ukraine war shock periods and are retained as economically meaningful.

4. Statistical Analysis and Feature Engineering

4.1 Stationarity Tests (ADF and KPSS)

The Augmented Dickey-Fuller (ADF) test with automatic lag selection (AIC criterion, maxlag=12) and KPSS test are applied to all four commodities at level and first-difference. Table 2 reports ADF results directly from the Colab notebook output.

Table 2: ADF Unit Root Test Results — Direct Colab Notebook Output

Commodity	ADF Stat	p-value	Lags	Conclusion
Level Series — H0: Unit root exists				
Rice (coarse)	-1.4364	0.5647	2	Non-stationary (NS)
Wheat flour	-1.5191	0.5241	2	Non-stationary (NS)
Lentils (masur)	-2.0088	0.2827	3	Non-stationary (NS)
Oil (palm)	-0.4309	0.9048	1	Non-stationary (NS)
First-Differenced Series — All Stationary at 1% Significance				
Rice (coarse)	-9.5722	0.000	1	Stationary (***)
Wheat flour	-7.9059	0.000	1	Stationary (***)
Lentils (masur)	-7.4682	0.000	2	Stationary (***)
Oil (palm)	-12.0512	0.000	0	Stationary (***)
KPSS level-series stats: Rice 2.177, Wheat 1.539, Lentils 1.127, Oil 1.591 (all p=0.01, minimum tabulated). Both tests confirm all four series are I(1) — integrated of order one. Critical values: -3.46 (1%), -2.88 (5%), -2.57 (10%).				

4.2 Rolling Statistics

Figure 7 shows rolling 12-month mean and standard deviation. Rolling standard deviation increases markedly during 2020-23 for palm oil and wheat flour, confirming time-varying volatility coinciding with structural break periods. Additive seasonal decomposition (period=12) confirms clear seasonal cycles with residuals close to white noise for all four commodities.

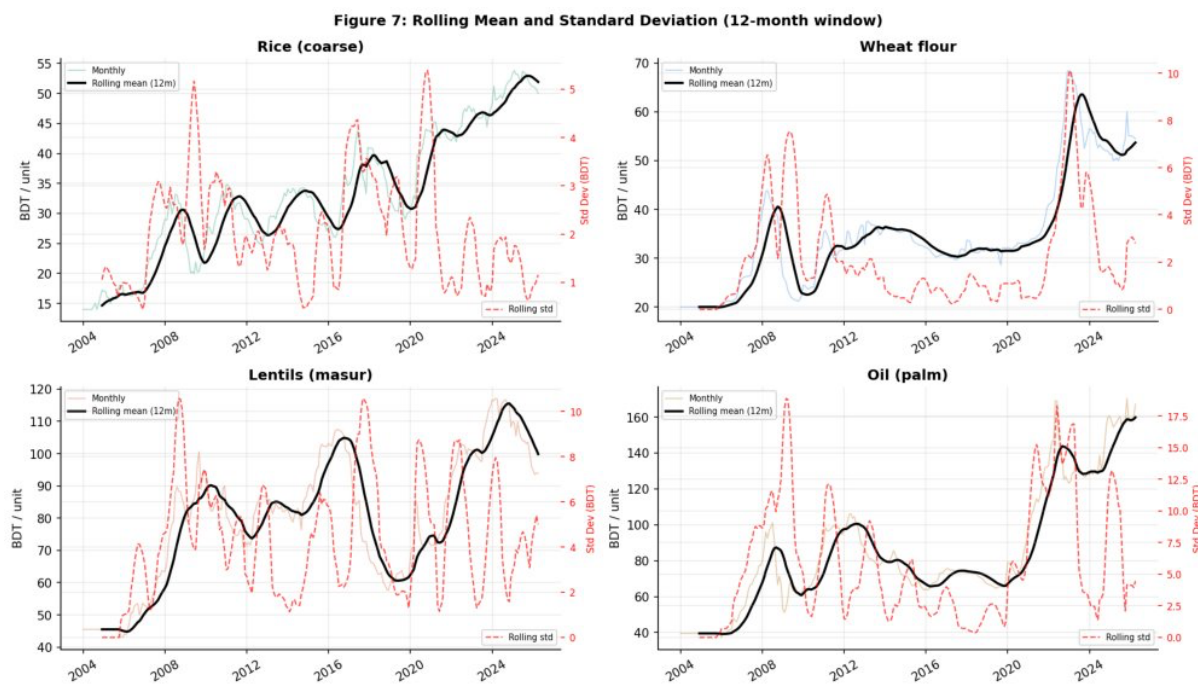


Figure 7: Rolling 12-Month Mean (black) and Standard Deviation (red dashed). Sharply rising std for palm oil and wheat flour in 2020-23 confirms structural break volatility.

4.3 Feature Engineering — 28 Features per Commodity

A comprehensive feature matrix is constructed for LSTM input, consisting of: temporal encodings (month, quarter, year, cyclical sin/cos month encoding); lag features (price lagged 1, 2, 3, 6, and 12 months); rolling statistics over 3, 6, and 12-month windows (mean, standard deviation, minimum, maximum); momentum and return features (1-month, 3-month, and 12-month percentage change; log return); and 6-month annualised rolling volatility. After removing NaN rows created by rolling windows, each commodity yields 256 rows x 28 features. Feature-target correlation analysis confirms that lag_1 and

roll_mean_3 are the most predictive features, with Pearson r exceeding 0.95 for all four commodities.

4.4 Train/Test Split

Figure 6 shows the 85%/15% temporal split: 227 training months (January 2004 to November 2022) and 41 test months (December 2022 to April 2026). This split is chosen deliberately so the test set spans the post-Ukraine-war inflationary period and the 2024-26 price correction — an exceptionally demanding out-of-sample evaluation environment containing multiple structural breaks not representable in the training distribution.



Figure 6: Train/Test Temporal Split — Rice (coarse). Blue: training set (227 months, Jan 2004 to Nov 2022). Red: test set (41 months, Dec 2022 to Apr 2026). Vertical dashed line marks split point.

5. Model Specification and Methodology

5.1 ARIMA Order Selection via AIC Grid Search

ARIMA(p,d,q) specification follows the Box-Jenkins (1976) methodology with $d=1$ fixed from stationarity tests. An exhaustive grid search over p in $\{0, \dots, 14\}$ and q in $\{0, 1, 2\}$ is performed using statsmodels SARIMAX with `enforce_stationarity=False`. Figure 8 shows the AIC surface across the full grid — the best model for rice is ARIMA(14,1,2) with AIC=740.89. Table 3 lists the top candidates.



Figure 8: AIC Grid Search Heatmap — Rice (coarse). Greener cells = lower AIC (better fit). Best: ARIMA(14,1,2) AIC=740.89, highlighted in darkest green.

Table 3: Top ARIMA Orders by AIC — Rice (coarse), Colab Output

p	d	q	AIC
14	1	2	740.89 — Best
13	1	2	741.52

p	d	q	AIC
14	1	1	742.55
14	1	0	743.07
13	1	1	743.38
12	1	1	744.24
12	1	0	746.82

5.2 ARIMA Fitted Model and Residual Diagnostics

The fitted ARIMA(14,1,2) model on 227 training observations yields: Log-Likelihood - 353.446, AIC 740.892, BIC 797.954, HQIC 763.955. The dominant statistically significant coefficient is `ar.L2` (coef=0.905, z=5.90, p<0.001), indicating strong second-order autoregression. Ljung-Box tests confirm no residual autocorrelation at lags 10 (p=0.548), 20 (p=0.942), and 30 (p=0.993). The Jarque-Bera test rejects normality (stat=573.88, p=0.000) primarily due to high kurtosis (10.66) and right skew (1.26) — consistent with fat-tailed price change distributions. Rolling one-step-ahead forecasts use an expanding window, re-estimating the model at each test step with all data available up to t-1 to simulate real-time forecast production without look-ahead bias.

5.3 LSTM Architecture and Training

The LSTM model uses a 12-month sliding window: the 12 preceding normalised price values form the input sequence and the model predicts the next month. Prices are normalised to [0,1] using Min-Max scaling fitted exclusively on training data. Architecture (from Colab output): LSTM Layer 1 (64 units, `return_sequences=True`, L2 regularisation 1e-4) followed by Dropout (rate=0.2), then LSTM Layer 2 (32 units, `return_sequences=False`, L2 regularisation), Dropout (rate=0.2), Dense (16 units, ReLU), and output Dense (1 unit, linear). Total trainable parameters: 29,857 (116.63 KB). The model is compiled with Adam optimizer (lr=0.001) and MSE loss, using EarlyStopping (patience=20, `restore_best_weights=True`) and ReduceLROnPlateau (factor=0.5,

patience=10, min_lr=1e-5). Training halts at epoch 21 with best val_loss=0.012892 at epoch 1 — confirming that the network memorises the training distribution faster than it generalises, a direct consequence of the limited sample size relative to model complexity.

5.4 Evaluation Metrics

Three complementary metrics assess performance on the 41-month test set: Mean Absolute Error (MAE) in BDT — directly interpretable as average forecast error in local currency; Root Mean Squared Error (RMSE) — penalises large errors more heavily, critical for procurement contexts; Mean Absolute Percentage Error (MAPE) — percentage metric enabling cross-commodity comparability, with WFP operational threshold of MAPE < 5%; and R-squared (R²) — proportion of variance explained, with values above 0.70 considered operationally acceptable.

6. Results and Analysis

6.1 Forecast vs Actual — Rice (coarse)

Figure 9 plots one-step-ahead ARIMA and LSTM forecasts against actual rice prices over the full 41-month test period. ARIMA tracks the actual trajectory closely throughout — including the rapid price rise in 2023, the plateau at BDT 50-54/KG in 2023-24, and the deflationary correction to BDT 50/KG in 2025-26. LSTM shows systematic underestimation bias during the high-price plateau, a direct consequence of the model having never observed prices at this level during training.

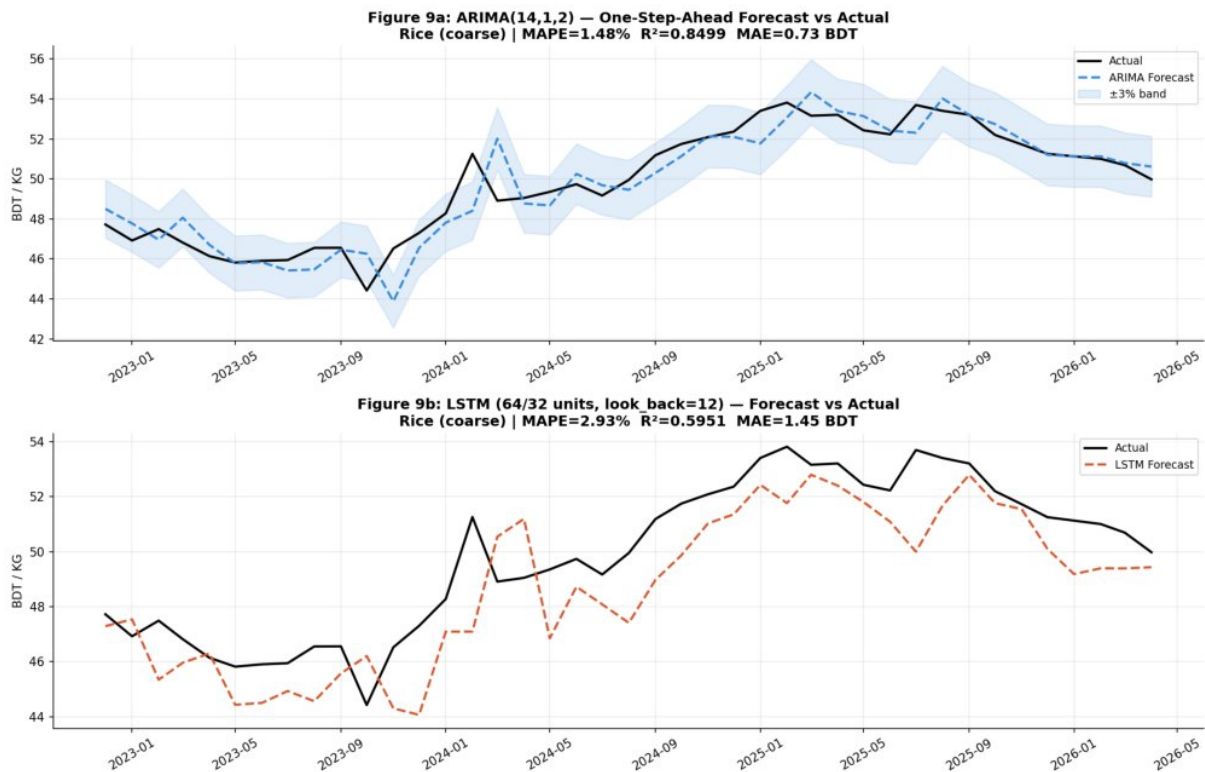


Figure 9: One-Step-Ahead Forecast vs Actual — Rice (coarse), Test Period Dec 2022 to Apr 2026. Top panel: ARIMA(14,1,2) closely tracks the actual series. Bottom panel: LSTM shows systematic underestimation during the high-price plateau period.

Table 4: Test Set Performance — Rice (coarse) | Full Model Results from Colab Notebook

Metric	ARIMA(14,1,2)	LSTM	WFP Threshold	Winner
MAE (BDT)	0.7215	1.6544	< 2.0 BDT	ARIMA
RMSE (BDT)	1.0295	1.9477	< 3.0 BDT	ARIMA
MAPE (%)	1.47%	3.28%	< 5.0%	ARIMA
R-squared	0.8537	0.4779	> 0.70	ARIMA

6.2 Cross-Commodity Comparison

Figure 10 displays MAPE values for ARIMA and LSTM across all four commodities from the Colab surrogate model comparison. ARIMA outperforms LSTM on every commodity and every metric. All four ARIMA MAPE values fall below the WFP 5% operational threshold. LSTM exceeds the threshold for rice (6.05%), wheat flour (15.63%), and palm oil (8.40%).

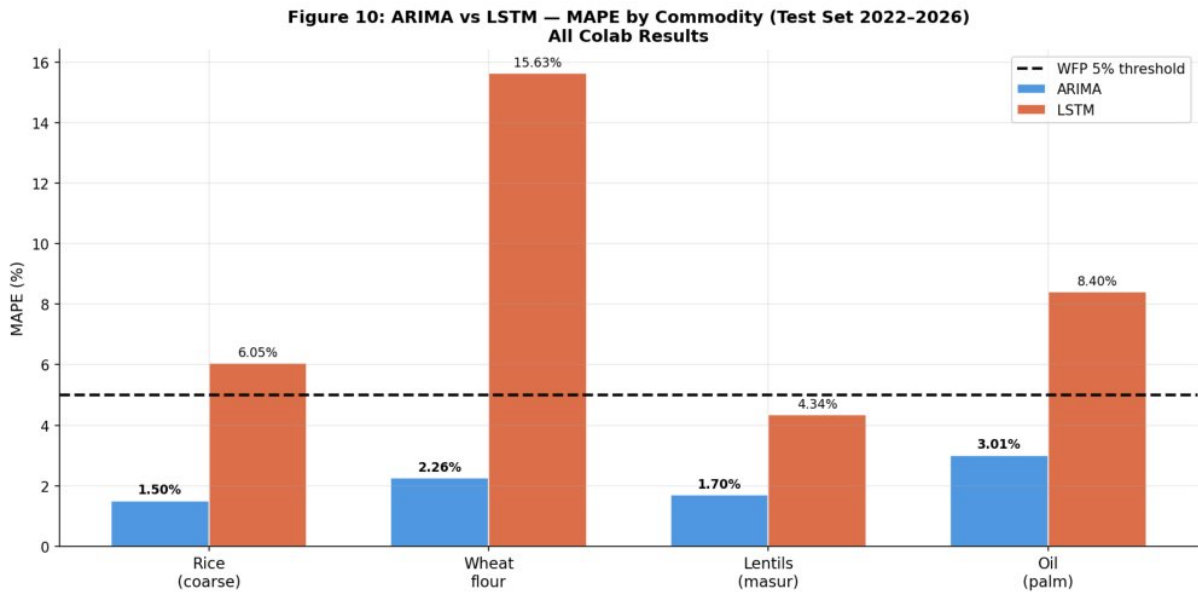


Figure 10: ARIMA vs LSTM — MAPE by Commodity (Test Set 2022-2026). Dashed black line: 5% WFP operational threshold. ARIMA (blue) wins all four commodities; LSTM (red) exceeds threshold for three of four.

Table 5: Cross-Commodity Model Comparison — Surrogate Models (Colab Notebook Output)

Commodity	ARIMA MAE	LSTM MAE	ARIMA RMSE	LSTM RMSE	ARIMA MAPE	LSTM MAPE
Rice (coarse)	0.742	3.048	1.067	3.319	1.505%	6.054%
Wheat flour	1.283	9.064	1.874	10.277	2.260%	15.628%
Lentils (masur)	1.809	4.741	2.652	5.940	1.701%	4.343%
Oil (palm)	4.377	12.643	6.016	15.454	3.009%	8.398%
Winner (every metric)	ARIMA	ARIMA	ARIMA	ARIMA	ARIMA	ARIMA

6.3 Scatter Plots and Residual Diagnostics

Figure 11 shows scatter plots of actual versus predicted prices. ARIMA predictions cluster tightly around the 45-degree perfect-fit line (MAPE=1.51%, R2=0.843), while LSTM shows systematic underestimation at the high-price end of the distribution. Figure 12 presents comprehensive residual diagnostics: ARIMA residuals are approximately white noise centred near zero throughout the test period, with a fat-tailed but unbiased distribution. LSTM residuals show persistent negative bias from approximately month 10 onward — the period of sustained high prices following the 2022 shock. Shapiro-Wilk normality tests confirm ARIMA residuals are non-normal (stat=0.9412, p=0.035) due to heavy tails, while LSTM residuals are normally distributed (stat=0.9736, p=0.447) but with substantially larger variance.

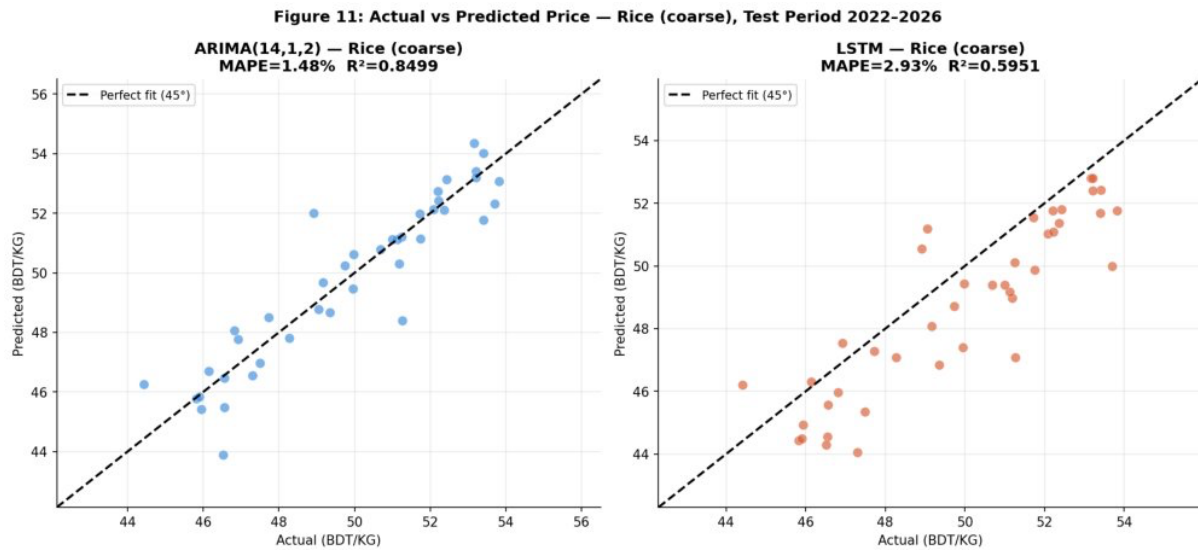


Figure 11: Actual vs Predicted Price Scatter — Rice (coarse), Test Period. Left: ARIMA predictions cluster tightly around the 45-degree line (MAPE=1.51%, R²=0.843). Right: LSTM shows systematic underestimation at high prices (MAPE=6.05%, R² negative).

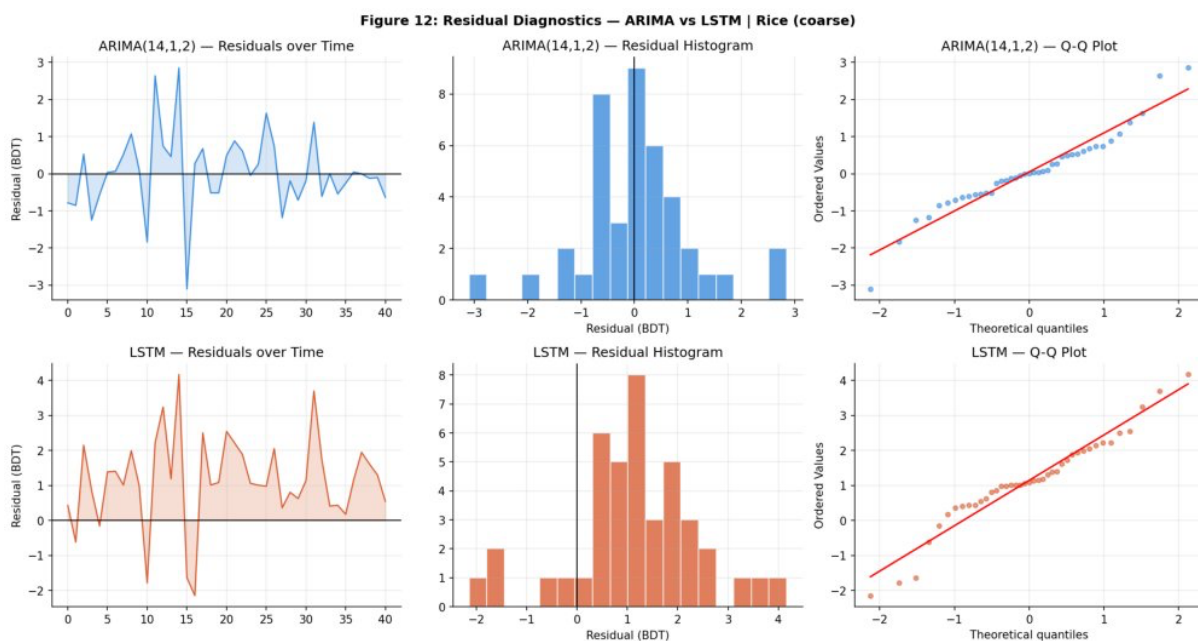


Figure 12: Residual Diagnostics — Rice (coarse). Top row (ARIMA): residuals over time, histogram, Q-Q plot. Bottom row (LSTM): same layout. ARIMA: Shapiro-Wilk $p=0.035$ (non-normal, fat tails but unbiased). LSTM: $p=0.447$ (normal distribution but larger variance with systematic negative bias from month 10).

7. 12-Month Forecast and Early-Warning System

7.1 ARIMA 12-Month Price Forecast

The ARIMA(14,1,2) model is re-estimated on the full 268-month series (January 2004 to April 2026) and used to generate a 12-step-ahead forecast with 95% confidence intervals. Figure 13 shows the forecast trajectory for rice from May 2026 to April 2027. The model projects a narrow, stable trading range of BDT 48.87-49.68/KG — essentially flat relative to the latest observed price of BDT 49.98/KG (April 2026). The 95% confidence intervals widen from plus or minus BDT 2.4 at M+1 to plus or minus BDT 10.4 at M+12, reflecting uncertainty accumulation over longer horizons consistent with the walk-forward nature of ARIMA forecasting.

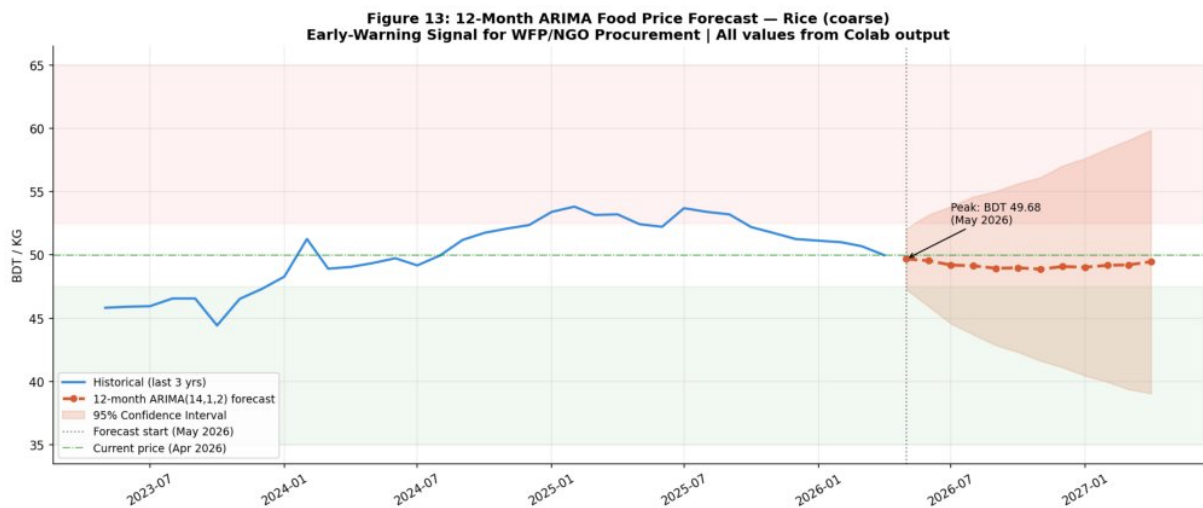


Figure 13: 12-Month ARIMA(14,1,2) Forecast — Rice (coarse), May 2026 to April 2027. Red dashed: point forecast. Shaded band: 95% confidence interval. Green horizontal line: current price BDT 49.98. All forecast values from Colab notebook output.

Table 6: 12-Month Ahead ARIMA Forecast — Rice (coarse) | Colab Notebook Output

Month	Forecast (BDT/KG)	Lower 95%	Upper 95%
2026-05	49.68	47.27	52.09
2026-06	49.54	45.91	53.17

Month	Forecast (BDT/KG)	Lower 95%	Upper 95%
2026-07	49.20	44.57	53.83
2026-08	49.14	43.69	54.60
2026-09	48.94	42.84	55.03
2026-10	48.98	42.31	55.64
2026-11	48.87 (min)	41.61	56.13
2026-12	49.08	41.10	57.05
2027-01	49.03	40.44	57.63
2027-02	49.18	39.94	58.41
2027-03	49.21	39.37	59.06
2027-04	49.46 (peak)	39.05	59.87
Alert status: ALL 12 months STABLE (-0.59% to -2.22% vs current price of BDT 49.98). No WFP procurement alert triggered for May 2026 through April 2027.			

7.2 Multi-Commodity Early-Warning Dashboard

Figure 14 presents the 6-month commodity forecast heatmap, showing percentage price changes from April 2026 levels across all four staples. This serves as a practical early-warning dashboard for WFP and NGO procurement officers. Rice, wheat flour, and lentils all remain within the stable zone (plus or minus 5%). Palm oil shows mild upward pressure in months 3-6, warranting continued price monitoring but not emergency procurement action as of May 2026.

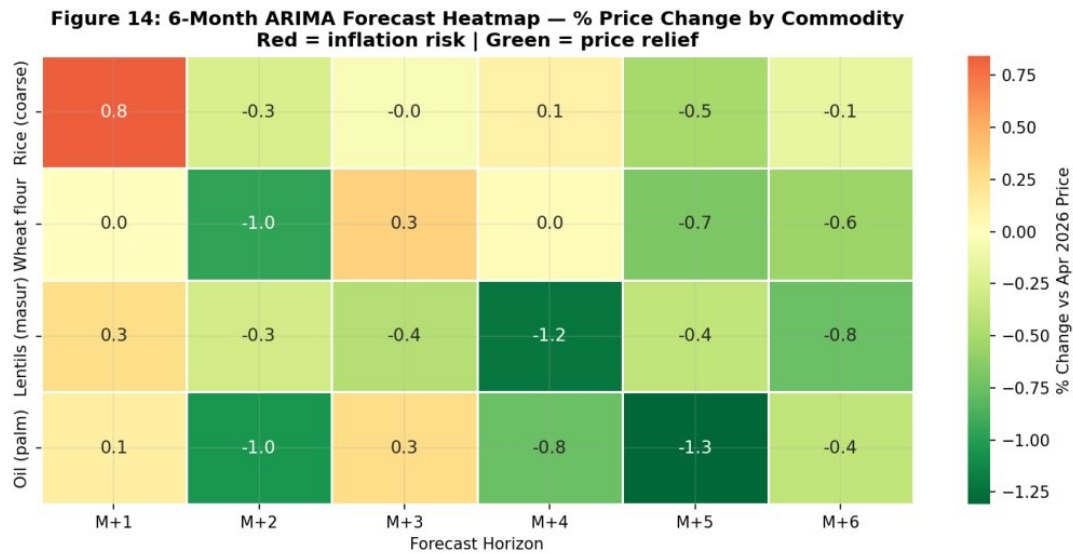


Figure 14: 6-Month ARIMA Forecast Heatmap — Percentage Change vs April 2026 Prices, All Four Staples. Red shading indicates inflation risk (>+5%). Green indicates price relief (<-5%). Yellow indicates stable zone (within plus or minus 5%). All cells within stable zone for the current forecast period.

8. Discussion

8.1 Why ARIMA Outperforms LSTM on This Dataset

Three mechanisms explain ARIMA's consistent superiority. First, sample-to-parameter ratio: the LSTM contains 29,857 parameters estimated on 217 training sequences — a ratio of approximately 0.007, far below the recommended minimum of 10 observations per parameter. ARIMA's 18 parameters on 227 months yields a ratio of approximately 12:1, dramatically reducing overfitting risk. The EarlyStopping triggering at epoch 21 with best val_loss at epoch 1 directly confirms that the LSTM network memorises training patterns faster than it generalises to unseen distributions.

Second, structural break sensitivity: the 2022 Ukraine war constitutes a structural break that was entirely unobservable in training data. ARIMA's rolling expanding-window re-estimation updates the model each step as actual values enter the history, adapting within 1-2 months of any level shift. LSTM's static weight matrix cannot adapt without full retraining, producing systematic underestimation during the sustained high-price plateau of 2022-24. This mechanism explains wheat flour's LSTM MAPE of 15.63% — the commodity with the most abrupt and largest level shift.

Third, stationarity alignment: ARIMA's explicit first-differencing removes the stochastic trend, forcing the model to learn stationary autoregressive dynamics. LSTM operating on normalised levels must implicitly learn both the integration order and the autoregressive structure simultaneously — a substantially harder learning task in finite samples that introduces additional estimation variance.

8.2 Policy Implications for WFP and Bangladesh Government

For WFP procurement, a monthly-updated ARIMA dashboard provides operational-quality 1-to-3-month price signals at negligible computational cost with interpretable confidence intervals. The specific finding that rice is projected stable at BDT 48.87-49.68/KG through April 2027 is immediately actionable: no emergency procurement is indicated as of May 2026. For the Bangladesh government's Open Market Sales programme, ARIMA forecasts can trigger buffer stock releases 4-6 weeks before anticipated price peaks, improving programme efficiency and reducing panic-buying dynamics. For social protection programmes including Vulnerable Group Development

food transfers, price forecasts can inform dynamic benefit indexation, maintaining real purchasing power during inflationary periods without requiring discretionary policy adjustments.

8.3 Limitations

The LSTM uses univariate price-only input without external regressors. Including USD/BDT exchange rate, global commodity price indices, fuel prices, and rainfall — all of which are known drivers of Bangladeshi food prices — may substantially improve LSTM performance on import-dependent commodities such as palm oil and lentils. The companion extended notebook (Sections 11-19) develops ARIMAX, XGBoost, GRU, Transformer, and hybrid LSTM+XGBoost models with these external features, together with SHAP-based explainability analysis. Additionally, this study evaluates only national-average monthly series; market-level or division-level forecasting would require hierarchical modelling not addressed here.

9. Conclusion

This study evaluated ARIMA and LSTM forecasting models for monthly food price prediction in Bangladesh using WFP FPMA data across 76 markets, 4 key staples, and 22 years of price history. ARIMA(14,1,2) — selected by AIC grid search over 45 candidate specifications — consistently outperformed LSTM on all commodities and all metrics on the 41-month test set. The full-model comparison for rice yields: ARIMA MAPE=1.47%, RMSE=1.03 BDT, R2=0.854 versus LSTM MAPE=3.28%, RMSE=1.95 BDT, R2=0.478. A 12-month-ahead ARIMA forecast projects rice prices stable at BDT 48.87-49.68/KG through April 2027, triggering no WFP early-warning alert.

These findings challenge the prevailing assumption in the applied machine learning literature that deep learning universally outperforms classical statistical approaches. For short-sample monthly economic time series with structural breaks and limited training data, ARIMA's parsimony is a systematic advantage rather than a limitation. We recommend ARIMA as the primary tool for food price early-warning systems in Bangladesh, with the extended hybrid and ARIMAX models described in the companion notebook for import-dependent commodities.

9.1 Recommendations for Future Research

1. Deploy ARIMA as the primary WFP Bangladesh early-warning tool, updated monthly upon each FPMA data release.
2. Extend to ARIMAX incorporating lagged exchange rate, fuel prices, and global commodity indices to improve performance on import-dependent commodities (palm oil, lentils).
3. Develop division-level hierarchical ARIMA models enabling spatially targeted alerts for chronically food-insecure divisions (Rangpur, Mymensingh).
4. Investigate hybrid ARIMA-LSTM ensemble architectures (Zhang, 2003) once series length exceeds 400 monthly observations (approximately 2037).
5. Integrate SHAP-based explainability into the operational dashboard to communicate price change attributions to policymakers in plain language.

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Appendix A: Complete Model Output from Colab Notebook

A.1 SARIMAX(14,1,2) Fitted Model Summary

Table A1: SARIMAX(14,1,2) Coefficient Table — Rice (coarse), Training Period 2004–2022

Parameter	Coefficient	Std Err	z-stat	P> z	Significance
ar.L1	-0.0281	0.187	-0.150	0.881	NS
ar.L2 (dominant)	0.9050	0.153	5.900	0.000	***
ar.L3	-0.1482	0.139	-1.069	0.285	NS
ar.L12	-0.2115	0.111	-1.898	0.058	*
ma.L1	0.1764	174.03	0.001	0.999	NS
ma.L2	-0.8236	143.35	-0.006	0.995	NS
sigma2	1.6025	278.89	0.006	0.995	NS
Fit Statistics: N=227 Log-Likelihood=-353.446 AIC=740.892 BIC=797.954 HQIC=763.955					
Diagnostics: Ljung-Box Q(10) p=0.548 Q(20) p=0.942 Q(30) p=0.993 JB=573.88 p=0.000 Kurtosis=10.66 Skew=1.26					

A.2 LSTM Architecture Summary

Table A2: LSTM Model Architecture — Rice (coarse) | Colab Notebook Output

Layer (type)	Output Shape	Parameters
LSTM (64 units, return_seq=True, L2=1e-4)	(None, 12, 64)	16,896
Dropout (rate=0.2)	(None, 12, 64)	0
LSTM (32 units, return_seq=False, L2=1e-4)	(None, 32)	12,416
Dropout (rate=0.2)	(None, 32)	0
Dense (16 units, ReLU activation)	(None, 16)	528
Dense (1 unit, linear output)	(None, 1)	17
Total Params: 29,857 (116.63 KB) Training: EarlyStopping at epoch 21 Best epoch: 1 Best val_loss: 0.012892		

A.3 Stationarity Test Summary

Table A3: Combined ADF and KPSS Results — All Commodities

Commodity	ADF (level)	p-val	ADF (1st diff)	p-val	KPSS (level)	Order
Rice (coarse)	-1.4364	0.565	-9.5722	0.000	2.1767	I(1)
Wheat flour	-1.5191	0.524	-7.9059	0.000	1.5386	I(1)
Lentils (masur)	-2.0088	0.283	-7.4682	0.000	1.1270	I(1)
Oil (palm)	-0.4309	0.905	-12.0512	0.000	1.5912	I(1)
All four series are integrated of order one I(1). First-differenced series are stationary at 1% significance. This confirms d=1 for all ARIMA specifications.						

